

TARBES LOURDES PYRENEES AP, France

WMO: 076210

Lat: 43.1881N

Lon: 0.0000E

Elev: 1260

StdP: 14.04

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	27.0	18.9	14.9	28.8	21.9	17.2	31.0	25.1	48.4	21.9	48.1	4.8	180	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.8	89.0	69.9	85.2	69.0	82.0	67.6	72.7	84.1	70.8	81.5	69.1	79.0	6.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	112.0	78.7	67.4	105.4	75.9	65.8	99.6	73.6	37.1	84.2	35.4	81.8	33.9	79.2	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	16.8	14.2	DB	18.9	95.9	3.4	3.2	16.4	98.2	14.4	100.0	12.5	101.8	10.0	104.1	(4)
(5)			WB	18.0	76.9	3.5	1.5	15.5	78.0	13.5	78.9	11.6	79.8	9.1	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	42.4	43.5	48.5	51.9	58.1	64.6	67.8	68.1	63.0	57.7	47.9	43.8
	DBStd	11.09	6.35	6.79	6.15	5.80	6.47	6.41	5.56	5.55	5.41	6.12	6.25	6.53
	HDD50	921	243	194	101	44	10	0	0	0	0	12	110	207
	HDD65	4096	701	602	512	395	228	85	35	30	102	236	512	658
	CDD50	2688	8	13	54	101	262	439	551	560	389	250	47	14
	CDD65	385	0	0	0	1	15	74	121	125	40	9	0	0
	CDH74	3640	0	1	3	31	214	773	1029	1067	407	114	1	0
	CDH80	1149	0	0	0	2	44	262	341	395	89	16	0	0
Wind	WSAvg	6.1	6.6	7.0	6.9	6.9	6.4	6.0	5.7	5.3	5.3	5.1	6.1	6.2
Precipitation	PrecAvg	44.8	4.2	3.5	3.4	4.6	4.6	3.4	3.0	2.7	3.0	3.5	5.2	3.8
	PrecMax	63.7	10.4	8.8	7.4	10.3	8.2	8.6	5.8	4.8	6.8	11.3	11.1	7.7
	PrecMin	36.1	0.4	0.7	0.5	1.7	2.0	0.9	1.1	0.6	1.1	1.4	0.9	0.4
	PrecStd	6.5	2.5	2.2	1.6	1.8	1.5	1.7	1.2	1.1	1.6	1.9	2.4	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	69.8	73.4	78.2	85.3	92.5	93.1	95.0	86.6	82.3	70.7	65.3
		MCWB	52.3	53.0	55.5	60.4	66.1	71.2	72.4	71.2	67.3	63.5	57.5	52.8
	2%	DB	59.3	63.1	68.9	72.9	79.5	86.5	87.5	89.0	82.5	77.1	66.4	61.1
		MCWB	50.0	50.3	54.7	58.0	64.0	68.5	70.3	70.3	66.1	62.4	55.4	51.3
	5%	DB	55.6	59.1	64.7	68.9	75.4	82.4	83.7	84.2	78.9	73.1	62.0	57.4
		MCWB	48.8	48.9	52.8	56.2	62.4	67.1	69.2	69.9	65.1	61.0	53.5	49.4
	10%	DB	52.5	54.8	60.7	64.5	71.3	78.0	79.9	79.9	75.1	69.4	57.7	54.0
		MCWB	47.8	47.5	51.0	54.3	60.5	65.8	68.0	68.3	63.6	59.8	52.0	48.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	54.3	58.2	62.0	68.6	74.4	75.3	75.2	71.0	66.4	60.1	55.4
		MCDB	59.7	63.8	68.4	74.1	80.1	85.9	87.4	87.1	81.1	75.8	67.5	61.5
	2%	WB	52.1	52.1	55.6	59.5	65.7	71.1	72.1	72.8	68.4	64.1	56.8	53.0
		MCDB	56.7	59.7	65.2	70.4	76.5	82.8	83.5	83.9	78.6	72.8	63.2	58.1
	5%	WB	50.2	50.2	53.7	57.2	63.5	68.6	70.2	70.9	66.3	62.3	54.8	51.1
		MCDB	54.3	56.4	62.8	66.8	73.6	79.3	80.9	81.1	75.5	70.5	60.0	55.8
	10%	WB	48.1	48.5	51.9	55.2	61.3	66.6	68.5	69.0	64.6	60.6	52.8	49.2
		MCDB	51.7	54.0	59.5	62.8	69.6	76.2	77.8	78.1	72.4	67.4	57.1	53.3
Mean Daily Temperature Range	5% DB	MDBR	16.0	17.2	18.8	18.6	18.3	18.8	18.4	18.8	19.5	19.1	16.4	16.7
		MCDBR	21.4	26.0	26.6	27.2	26.3	27.9	27.3	27.5	26.3	25.5	23.2	22.4
		MCWBR	14.0	15.5	15.0	14.7	13.8	13.3	12.9	12.8	12.7	13.5	14.0	14.2
	5% WB	MCDBR	18.6	21.3	22.8	23.9	23.6	23.8	23.2	23.5	22.8	22.0	19.4	18.6
MCWBR		13.3	14.0	13.6	13.8	13.0	12.4	12.0	11.9	11.9	12.4	12.9	12.9	
Clear-Sky Solar Irradiance	taub	0.299	0.315	0.353	0.377	0.389	0.402	0.393	0.394	0.364	0.344	0.315	0.297	
	taud	2.502	2.447	2.363	2.312	2.321	2.318	2.347	2.352	2.425	2.484	2.508	2.512	
	Ebn at Noon	268	281	281	282	280	276	277	273	274	266	259	258	
	Edh at Noon	22	28	34	39	40	40	38	37	32	27	22	20	
All-Sky Solar Radiation	RadAvg	513	798	1178	1477	1710	1936	1955	1735	1401	936	586	469	
	RadStd	49	93	125	151	138	123	116	105	67	78	66	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page